



Electrochemical lithium intercalation behavior of pristine and milled hexagonal boron nitride

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ABSTRACT

Hexagonal boron nitride (h-BN) has been explored for lithium intercalation due to its similar crystal structure to graphite, a well-known anode material for Li-ion secondary batteries. Lithium h-BN intercalation compounds (Li-BNICS) have been successfully synthesized through heat treatment. In this study, potential physical properties related to electrical conductivity were investigated using pristine and milled h-BN, which is further mixed/combined with milled graphite. The electrochemical properties were characterized by cyclic voltammetry (CV) and galvanostatic cycling with potential limitation (GCPL). The chemical potential of h-BN was estimated about 1.0 V versus Li/Li⁺ and a two-phase reaction was suggested. Accordingly, Li-BNIC is more stable than Li-GICs (lithium graphite intercalation compounds) in terms of thermal stability.

1. Introduction

Graphite is widely used as anode materials for lithium-ion secondary batteries (LIBs), where it can intercalate and deintercalate lithium ions based on the layered-structure characteristics [1,2]. In a fully charged state, LiC₆ is formed, and its theoretical capacity is 372 mAh/g with a potential below 0.3 V versus Li/Li⁺. During the charge/discharge reaction, Li concentrations in graphite are changed, exhibiting various repeated interlayer distances (*I_c*) of Li-GICs (lithium graphite intercalation compounds) [2]. The *I_c*s can be described as the following equation [3,4]:

$$I_c = 0.37 \text{ nm} + (n - 1) \times 0.335 \text{ nm} \quad (1)$$

where 0.37 and 0.335 nm are the interlayer distances between graphene planes with and without Li intercalation, respectively, and *n* is the number of graphene planes between two intercalated layers. The Li-GICs are then described as the so-called *n*-stage phases; 1st stage phase (LiC₆), 2nd stage phase (LiC₁₂ [LiC_{6n}] and LiC₁₈ [LiC_{9n}]), 3rd stage phase (LiC₂₇ [LiC_{9n}]) and so on [2,5]. Recently, in order to increase the capacity for the batteries, many carbonaceous materials have been studied including milled graphite [6–9], carbon nanotubes [10–13], superdense Li-GICs [4,14,15] and so on. On the whole, the capacities are increased, however, there are problems such as larger potential hysteresis [7] and smaller charge-transfer resistance [9]. C. Natarajan et al. suggested a short milling duration (< 5 h) was effective

for reducing the reversible capacity and increasing the irreversible capacity as “the extended milling destroys surface and increases the basal plane surface area” [8].

As the atomic structure of hexagonal boron nitride (h-BN, *a* = 0.2505 nm, *c* = 0.666 nm [16]) is similar to graphite (*a* = 0.245 nm, *c* = 0.670 nm), lithium intercalation into h-BN has also been studied, although it is an insulator unlike graphite. The lithium boron nitride intercalation compounds (Li-BNICS) have been synthesized through heat treatment [17,18] and sequential processes of ball milling and heat treatment [19]. The exact atomic structure and Li concentration of Li-BNIC have not been reported although the layered structure of Li-BNIC has been suggested to exhibit “1 L model”, but not “2 L model”. The 1 L model involves “a structure without two-layer stacking periodicity”, while the 2 L model “two-layer stacking periodicity of pristine h-BN” [18]. As this suggests, Li could be intercalated into defective BN. Similar *I_c* values for Li-BNIC have been reported, e.g. 0.37599 nm [18] and 0.3752 nm [19], which was unchanged with different ratios of Li/BN [19]. Variations of *I_c* between pristine and intercalated graphite (LiC₆: 0.37 nm) are quite similar to h-BN. Both are thus comparable, and the detailed structure of Li-BNIC is worth investigating.

Meanwhile, Li-GICs are not stable at higher temperatures and deintercalation may occur, although there is no consensus on the temperature [20–23]. In order to produce Li-GICs by heat treatment a temperature below 400 °C must be selected to suppress synthesis of

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Li_2C_2 (it is not an intercalation compound) [24]. On the other hand, Li-BNIC is more stable than Li-GICs in terms of thermal stability, where it has been reported to be synthesized at higher temperatures (650 [19] and 1250 °C [17]) and it has been discussed in Li-BN-graphite system from a sequential process of ball milling and heat treatment by Kim et al. [25].

In spite of the potential and successful production of Li-BNICs, electrochemical lithiation of h-BN has not been reported by now. Electrochemical properties of intercalation compounds of h-BN are extremely important for investigating its possibility of application in terms of Li-ion battery. Therefore, the purpose of this study is to investigate the potential physical/chemical properties relating to electrical conductivity using pristine and milled h-BN with/without graphite.

2. Methods

2.1. Ball mill

Graphite and h-BN were ball-milled using a vibrational ball mill machine (NISSIN GIKEN Corporation, Super Misuni NEV-MA8). Three kinds of starting materials, which are graphite, h-BN and their mixture (1 : 8 on weight ratio), were prepared with total weight of 3 g and they were put in an AISI 306 stainless vessel with nine stainless balls (about 3.58 g per a ball) under Ar atmosphere. Then, ball milling has been conducted for 2.5 h with a vibrational frequency of 710.

Milled samples were characterized by XRD (X-ray diffractometry; $\text{Cu-K}\alpha$ radiation) to investigate varying defective properties. From XRD peaks, crystallite sizes (L) for milled samples were calculated using a Scherrer equation [26] as follows:

$$L = \frac{K \lambda}{\beta \cos \theta} \quad (2)$$

where, K is a shape factor (in the parallel layer group, $K = 0.9$ for $00l$ diffraction (L_c) and $K = 1.84$ for hkl diffraction (L_a) [27,28]), λ is the X-ray wavelength ($\lambda = 0.154060$ nm in the case of $\text{Cu-K}\alpha$), β is FWHM (full-width at half maximum) in radians, and θ is the Bragg angles. Specific surface areas of samples were estimated using BET (FlowSorb III 2305, Micromeritics Instrument Corporation), which is the method based on the physical adsorption of gas, developed by the Brunauer, Emmet and Teller (BET) [29]. Here, the samples were purified by degassing at 180 °C for 30 min, followed by the BET measurement using the mixed gas of N_2 30% and He 70%. The FE-SEM (Field emission scanning electron microscopy, Hitachi High-Technologies Corporation, SU6600) was conducted for observing particle sizes and surface states of samples. In order to investigate reaction between h-BN and graphite by milling, XPS (X-ray photoelectron spectroscopy) with Al $\text{K}\alpha$ source operating at 10 kV and 20 mA was conducted. The pass-energy was 20 eV with 0.1 eV / 200 ms steps, and the targets were etched with 300 V for 0, 30 and 180 s.

2.2. Working electrode (WE) preparation

A typical construction of beaker cells used here is illustrated in Figure 2a.

Table 1 shows detailed information about prepared working electrodes (WEs). There are six kinds of WEs: pristine graphite (Pure C), pristine h-BN (Pure BN), milled graphite (Milled C), milled h-BN (Milled BN), mixture of Milled C and Milled BN with a weight ratio of 1: 8 (Milled BN8C1-s), and milled-together graphite and h-BN with the same ratio (Milled BN8C1-t). The prepared six kinds of powders were mixed with the binder, which consists of 5.34 wt% polyvinylidene fluoride (PVDF) mixed with an N-Methyl-2-pyrrolidone (NMP) solvent, and coated on a Cu foil and dried at 120 °C for 12 h in vacuum.

Table 1

Constituted arrangement of working electrodes.

No.		Active and/or conductive material		Binder	Weight (mg)	Named
		h-BN	Graphite			
1	Category	–	Pure graphite	PVDF	1.95 ± 0.005	Pure C
	wt%	–	90	10		
2	Category	Pure BN	–	PVDF	1.3 ± 0.005	Pure BN
	wt%	90	–	10		
3	Category	–	2.5 h-milled graphite	PVDF	1.3 ± 0.005	Milled C
	wt%	–	80	20		
4	Category	2.5 h-milled BN	–	PVDF	0.95 ± 0.005	Milled BN
	wt%	90	–	10		
5	Category	2.5 h-milled BN	2.5 h-milled graphite	PVDF	1.2 ± 0.005	Milled BN8C1-s
	wt%	80	10	10		
6	Category	2.5 h-milled BN + graphite	–	PVDF	0.9 ± 0.005	Milled BN8C1-t
	wt%	80	10	10		

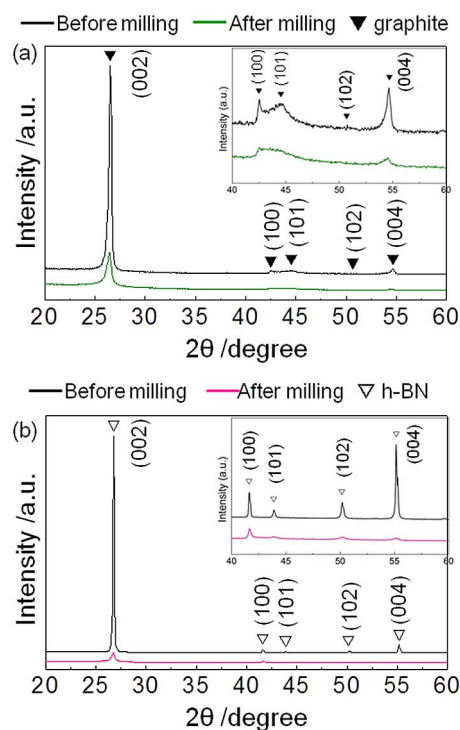


Fig. 1. XRD patterns of pristine (Before milling) and 2.5 h-milled (After milling) powders: (a) graphite and (b) h-BN, with expanded inset figures from 40 to 60 degrees.

Table 2

Crystallite sizes (L_c and L_a) and specific surface areas of the pristine (0 h-milled) and 2.5 h-milled samples for h-BN and graphite.

	Graphite		h-BN	
	0 h	2.5 h	0 h	2.5 h
L_c (nm)	31	17	60	20
L_a (nm)	50	–	155	70
Specific surface area (m^2/g)	9.7	259.8	1.2	258.7

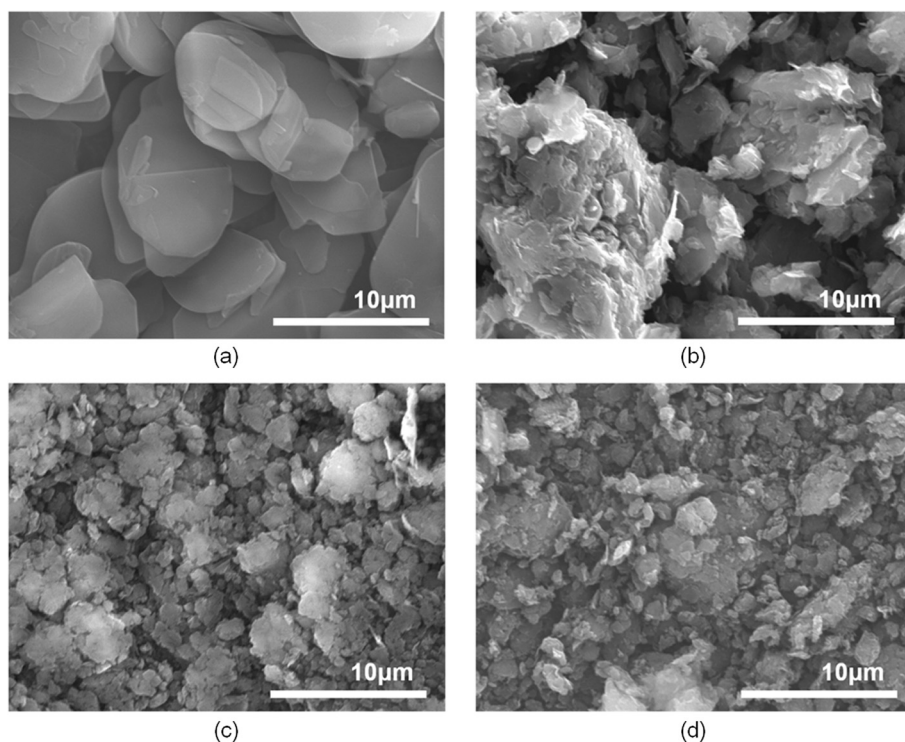


Fig. 2. SEM images for (a) pristine h-BN, (b) pristine graphite, (c) 2.5 h-milled h-BN and (d) 2.5-milled graphite.

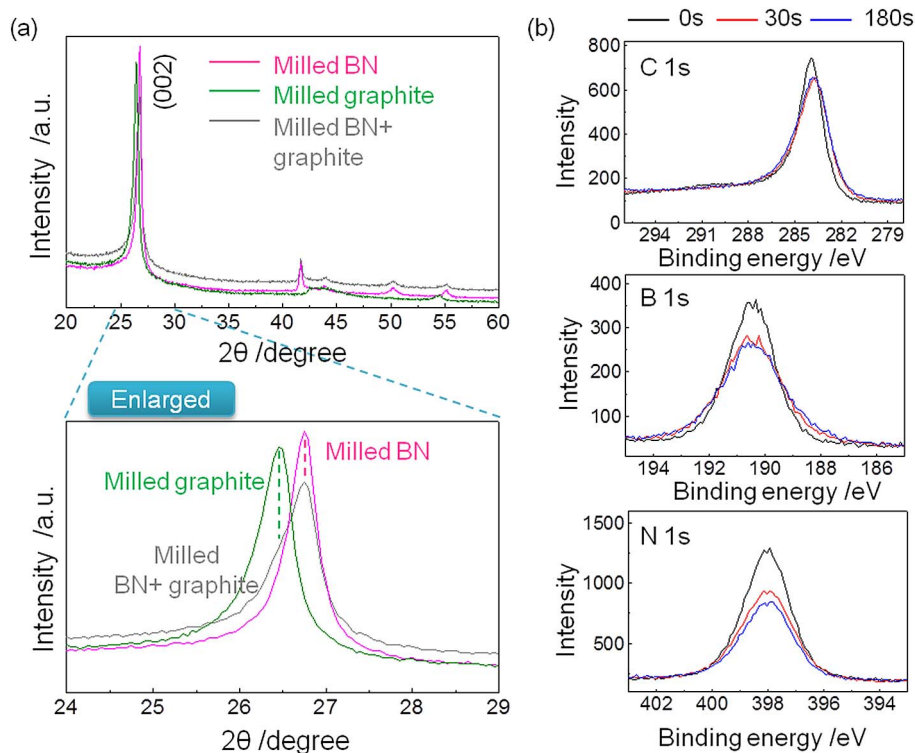


Fig. 3. (a) XRD patterns of 2.5 h-milled h-BN, graphite, and together (h-BN + graphite with weight ratio of 8 : 1) and below figure shows enlargement from 24 to 29 degrees. (b) XPS C 1s, B 1s and N 1s profiles of milled-together h-BN and graphite with etching times of 0, 30 and 180 s.

2.3. Electrochemical performance

The electrochemical performances were evaluated using a beaker cell with three electrodes in ethylene carbonate/dimethyl carbonate (EC-DMC 1:2 by vol.) containing 1 M LiPF₆. The counter and reference electrodes are lithium ribbons. Assembling the cell and all electrochemical measurements were conducted in a glove box filled with Ar gas with moisture and oxygen levels below 2 ppm. Electrochemical properties were measured by cyclic voltammetry (CV) with 1 mV/s in a

range of 0–2 V and galvanostatic cycling with potential limitation (GCPL) using a potentiostat/galvanostat (Biologic Science Instruments, VMP3).

2.4. Theoretical capacities for h-BN

Theoretical capacities for an electrochemical reaction are typically calculated by Faraday's 2nd law of electrolysis as follows:

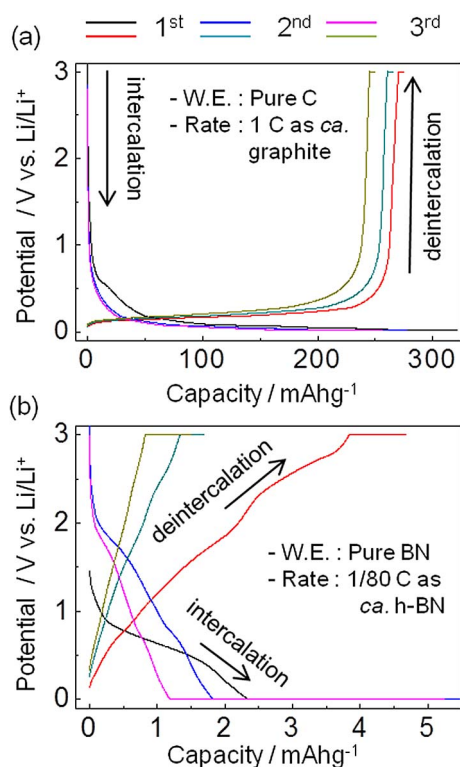


Fig. 4. Galvanostatic cycles for (a) pristine graphite and (b) pristine h-BN.

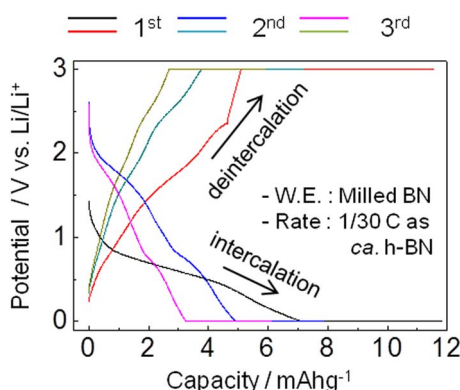


Fig. 5. Galvanostatic cycles for 2.5 h-milled h-BN.

$$Q = \frac{xF}{M} \quad (3)$$

where Q is a theoretical specific capacity (mAh/g), x is the (max.) number of Li moles for reaction involved, M is a molar mass and F is Faraday constant ($F = 96,485 \text{ C (A} \cdot \text{s)} = 26,800 \text{ mAh}$). The capacity of graphite was calculated as 372 mAh/g based on the following reaction:



On the other hand, in the case of BN, the reaction(s) involved and the Li concentration(s) have not been reported. Therefore, a hypothetical reaction similar to (4) for producing LiC_6 was tentatively written as follows:



From Eq. (5), the calculated capacity for BN is 360 mAh/g and this estimated capacity value will be used for calculating C-rate.

3. Results and discussion

3.1. Milling characteristics

Fig. 1 shows XRD patterns of graphite (a) and h-BN (b) before and after milling (Milling was separately performed for 2.5 h.). After milling, high-crystallinity structures in pristine graphite and BN were destroyed and their XRD peaks were broadened with decreased intensities.

Table 2 shows the estimated crystallite sizes, using the Scherrer equation from XRD peaks in Fig. 1, and the BET specific surface areas, of pristine and 2.5 h-milled samples for graphite and h-BN. Diffraction contributions from $\text{Cu-K}\alpha_2$ radiation are subtracted before calculation and the L_c and L_a values are calculated from (002) and (100) diffraction peaks, respectively. For milled graphite, only L_c was shown, as the (100) peak was too broad, as seen in Fig. 1 (a), to provide a credible L_a value. The L_c values are always smaller than L_a values, characteristic to the cleavage mechanism involved in the structure. The milled graphite and h-BN show smaller crystallite sizes than pristine materials. Although crystallite size (L_c) of milled h-BN is still larger than that of milled graphite, the reduction rate for h-BN before and after milling is larger than for graphite. It is supposed that ball milling has greater influences on h-BN for shorter milling durations than on graphite, possibly due to the differences in hardness and chemical bonding.

The structural characteristics of the milled BN have been reported by several researchers [30–36]. The processes for deformation of h-BN structure during ball milling are accordingly divided into three steps as follows: For shorter milling durations (for 1 h using a planetary ball mill [32] or ~5 h using a high-energy vibrational ball mill [35]), mainly the crystallite size of h-BN was decreased with small amount of stacking faults due to cleavage along the basal plane. For intermediate milling durations (for 2 h [32] or 7–9 h [35]), both slower reduction of crystallite size and formation of a number of defects such as simultaneous shearing of lattice planes, inhomogeneous strains, twin faults and stacking faults occurred. Finally, the nanocrystalline powders with amorphous, turbostratic or both structures were formed through accumulation of above defects.

Fig. 2 shows the SEM images for pristine powders [(a) h-BN, (b) graphite] and 2.5 h-milled samples [(c) h-BN, (d) graphite]. The particle sizes (or agglomerates) of pristine h-BN and graphite are much larger than milled h-BN and graphite, respectively, where the change appears more significant for the former. Also, the specific surface area of 2.5 h-milled h-BN is significantly increased from 1.2 to 258.7 m^2/g , as seen in Table 2, consistent with above SEM observations. The variation of surface area for milled h-BN has also been reported by Streltskii et al., where it has increased from 12 (0 h milling) to 400 m^2/g [36]. Thus, the set milling duration of 2.5 h in this study for h-BN may correspond to the initial step of deformation processes, involving large reduction of both crystallite size and particle size, which leads to the significant increase in the specific surface area.

The XRD patterns for various milled samples (Milled C, Milled BN, Milled BN8C1-t) are shown in Fig. 3 (a) on the left and the XPS (C 1 s, B 1 s and N 1 s) profiles for Milled BN8C1-t with etching times of 0, 30 and 180 s are shown in Fig. 3 (b) on the right. The XRD patterns of Milled C and Milled BN are duplicated in Fig. 1 and 3 for comparison. It could be concluded that BN and graphite did not react after milling because: first, as seen in an enlarged figure of Fig. 3 (a), a shoulder XRD peak for milled graphite can be seen at the lower angle side, and second, in Fig. 3 (b) the XPS spectra are mostly unchanged (except for the intensities) for graphite (at around 284.0 eV of C 1 s) and h-BN (at around 190.5 eV of B 1 s and 398.0 eV of N 1 s [37]) regardless of etching times.

3.2. Electrochemical characteristics

The galvanostatic cycles with potential limitation (GCPL) of pristine

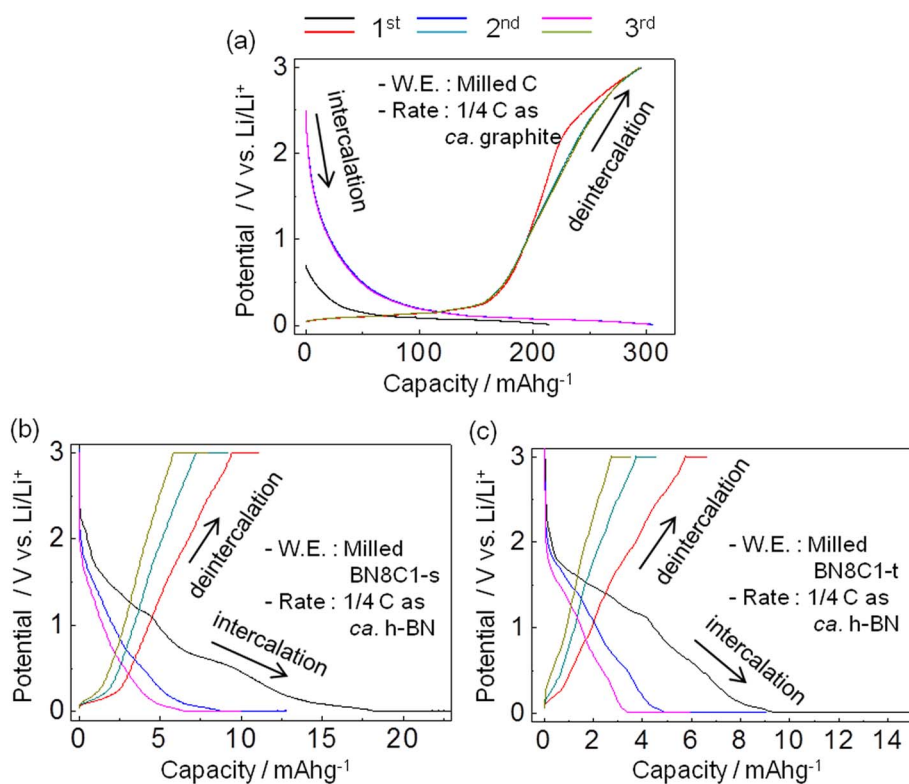


Fig. 6. Galvanostatic cycles for (a) 2.5 h-milled graphite, (b) mixture of 2.5 h-milled h-BN and 2.5 h-milled graphite with a weight ratio of 8 : 1 and (c) 2.5 h-milled-together h-BN and graphite with the same weight ratio.

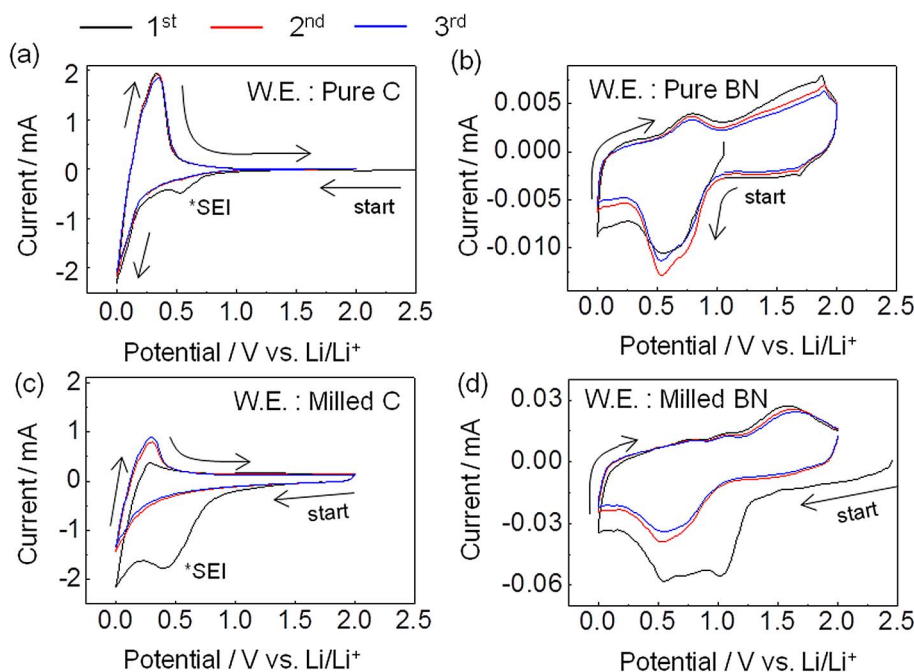


Fig. 7. Cyclic voltammograms for pristine (a) graphite and (b) h-BN and 2.5 h-milled ((c) graphite, (d) h-BN) materials.

graphite and h-BN are shown in Fig. 4. The GCPLs were conducted on Pure C and Pure BN: for the former using the 1 C-rate calculated by the theoretical capacity of graphite (372 mAh/g), while for the latter 1/80 C with the capacity of h-BN (360 mAh/g). In the case of BN, a very slow rate of 1/80 C was selected, as it was difficult to measure GCPL with 1 C that was used for pristine graphite. Fig. 4 (a) shows typical galvanostatic cycles of graphite. On the other hand, significantly small capacities were obtained for Pure BN even though the C-rate was very small, as seen in Fig. 4 (b). Accordingly, it is supposed that Li intercalation into pristine h-BN is extremely difficult through an electrochemical method.

The GCPL of only-milled h-BN was also conducted in order to investigate the effect of ball milling using the C-rate of 1/30 C. Although the rate is faster than that used for Pure BN, as shown in Fig. 5, the capacity of Milled BN is higher than that of Pure BN. This might be due to the increased surface area and/or the formation of defective h-BN through milling, as explained earlier (Fig. 1, Fig. 2 and Table 2). The enhanced capacity, however, still falls short of the capacity exhibited by graphite.

In order to improve the electrical conductivity of Milled BN, 2.5 h-milled graphite was added to 2.5-milled BN by weight ratio of 1 : 8 (Milled BN8C1-s), and the GCPL measurement was conducted using the

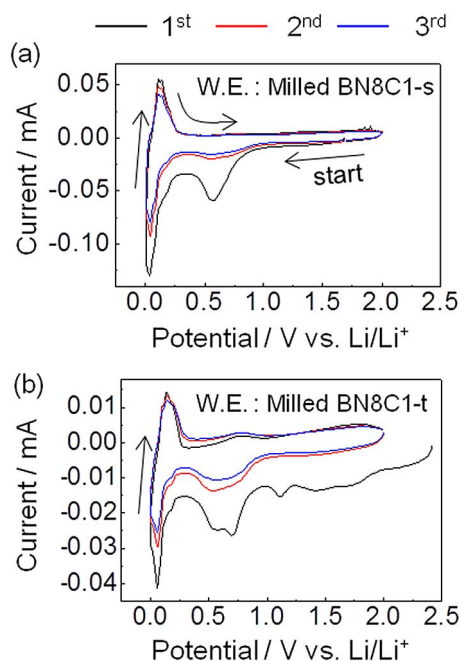


Fig. 8. Cyclic voltammograms for (a) mixture of 2.5 h-milled h-BN (Milled BN) and graphite (Milled C) by weight ratio of 8 : 1 and (b) 2.5 h-milled-together h-BN and graphite by the same ratio.

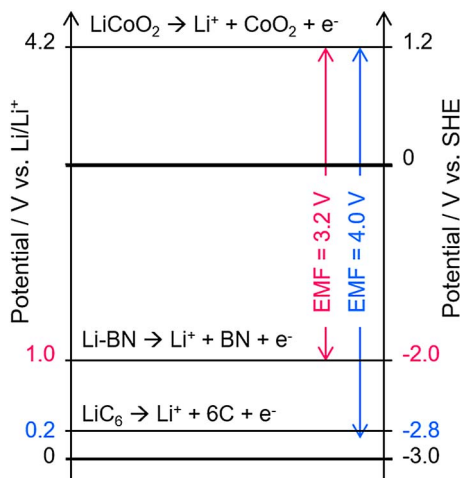


Fig. 9. Suggested chemical potentials of h-BN compared with graphite and CoO_2 .

rate of 1/4 C. The milled graphite (Milled C) and milled-together h-BN and graphite by the same ratio (Milled BN8C1-t) were also used for GCPL measurement with the same C-rate (1/4 C) to confirm an influence of milling on electrochemical properties. Fig. 6 shows the result of GCPL for (a) Milled C, (b) Milled BN8C1-s, and (c) Milled BN8C1-t. The Milled C shows a rather stable charge/discharge reaction behavior, where the second and third cycle shapes are almost similar, as seen in Fig. 6 (a), in contrast with Pure C (Fig. 4 (a)). The capacities of Milled BN8C1-s and Milled BN8C1-t are higher than Pure BN and Milled BN in Fig. 4 (b) and Fig. 5, respectively, though it falls short of Pure C (Fig. 4 (a)) and Milled C (Fig. 6 (a)). The shapes of each curve for Milled BN8C1-s and Milled BN8C1-t exhibit a mixed behavior of h-BN and graphite. In charging reactions, Li intercalation into h-BN firstly occurs, as seen in the rapid curve shape, while the smooth curve shape should represent intercalation of Li into graphite. On the other hand, in the discharge reaction, Li was first deintercalated from graphite until about 2.5 mAh/g for Milled BN8C1-s, or about 1 mAh/g for Milled BN8C1-t at the first cycle, followed by deintercalation from h-BN. Upon comparing

Milled BN8C1-s with Milled BN8C1-t, the capacities of the latter are smaller and more similar to Milled BN than the former.

All GCPL results show that lithium intercalation/deintercalation into BN is rather fast but in a small quantity. The difficulty of an intercalation process involving only a small amount of Li is probably ascribed to the reaction occurring only near the surface region of BN through the electrochemical method, besides that Li atoms diffuse very slowly into BN.

Fig. 7 shows the cyclic voltammograms of (a) Pure C, (b) Pure BN, (c) Milled C and (d) Milled BN. The Pure C and Milled C samples exhibit the typical cyclic voltammograms of graphite: the lithium intercalation reaction starts at about 0.2 V vs. Li/Li^+ and deintercalation finishes at about 0.3 V vs. Li/Li^+ [4]. A solid-electrolyte interphase (SEI) should form for both cases at around 0.5 V vs. Li/Li^+ , which does not exhibit from the second cycle. The equilibrium intercalation potential of h-BN is around 1.0 V vs. Li/Li^+ and it was also obtained in the milled h-BN. The CV shapes of Pure and Milled BN indicate a two-phase reaction, where un lithiated and lithiated phases coexist during the charge/discharge process [38–40]. For the Milled BN at the 1st cycle, two kinds of shoulders about 1.1 V and 0.5 V vs. Li/Li^+ were obtained because the surface of milled h-BN is metastable.

Fig. 8 shows the cyclic voltammograms of (a) Milled BN8C1-s and (b) Milled BN8C1-t. In both samples, except for the 1st cycle, two kinds of lithium intercalation reaction for h-BN and graphite were obtained at 1.0 V and 0.2 V, respectively, which correspond to each equilibrium intercalation potential. This mixture of electrochemical behaviors of cyclic voltammograms is consistent with the galvanostatic cycles of Milled BN8C1-s and -t (Fig. 6 (b) and (c)).

Fig. 9 shows estimated chemical potentials, vs. Li/Li^+ and vs. SHE (standard hydrogen electrode), of h-BN and graphite. In order to calculate EMF (electromotive force) from these CV results, LiCoO_2 (or CoO_2), which is a typical cathode material of Li-ion batteries, is also shown for comparison. The diagram can be explained as that Li-BNics are more stable than Li-GICs in terms of thermodynamics and Li intercalation into h-BN occurs more quickly than into graphite. It is consistent with the thermal stability of Li-GICs and Li-BNics as mentioned in the Introduction section and as report by Kim et al. [25]. This result implies that h-BN can be considered as a potential battery material for high-temperature application. In this study, firstly the electrochemical properties of h-BN at only room temperature region were investigated, but the study on its feasibility for higher temperature is required in further investigation.

As a whole, milled h-BN is better than pristine one in terms of the electrochemical properties, although the mixtures with graphite are unaffected by ball milling. It implies that there is a possibility of improving the electrochemical performance of defective h-BN. Further detailed study on Li concentration in Li-BNics is required for determining the specific capacity.

4. Conclusion

Electrochemical experiments of CV and GCPL under an ambient condition were conducted on pristine graphite, pristine h-BN, and 2.5 h-milled various counterparts. The chemical potential of graphite is found 0.2 V vs. Li/Li^+ , while of h-BN 1.0 V vs. Li/Li^+ , regardless of pristine or milled. Thus, Li-BNics are more stable than Li-GICs in terms of thermodynamics under an ambient condition. The Li intercalation occurs on the surface of BN, and then Li atoms/ions are very slowly diffused inside BN particles. Accordingly, Li intercalation into BN is more difficult than into graphite through an electrochemical process although milled BN exhibits higher capacities than pristine BN (due to larger surface area of the milled BN). Lastly, CV curves indicate a two-phase reaction.

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